

OpenSHMEM Message Aggregation Pass for MLIR

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1 Introduction

The OpenSHMEM Message Aggregation Pass is a compiler optimization implemented within the Multi-Level Intermediate Representation (MLIR) framework that analyzes and optimizes OpenSHMEM Remote Memory Access (RMA) operations. The pass identifies opportunities to coalesce multiple small communication operations into fewer, larger operations, thereby reducing communication overhead and improving application performance.

2 Pass Architecture

The message aggregation pass employs a multi-phase analysis and transformation approach consisting of four main components:

2.1 Memory Access Analysis

The `MemoryAccessAnalyzer` component extracts memory access patterns from OpenSHMEM operations, creating a unified representation that captures:

- Operation type (PUT, GET) and variant (MEMORY, TYPED, SIZED, POINT)
- Target processing element (PE) and context information
- Memory references for source and destination
- Transfer sizes and blocking behavior
- Element sizes for typed operations

2.2 Memory Layout Analysis

The `MemoryLayoutAnalyzer` performs sophisticated memory region analysis to determine:

- Contiguity of memory accesses through offset calculation
- Stride patterns for regular non-contiguous accesses
- Base address extraction from complex memory reference chains
- Support for `memref.subview` and `memref.view` operations

2.3 Communication Pattern Detection

The `CommunicationPatternDetector` groups operations based on compatibility criteria:

- Same target PE and context
- Identical blocking behavior (blocking vs. non-blocking)
- Compatible operation types and variants
- Matching element sizes for sized operations

2.4 Transformation Engine

The `TransformationEngine` applies coalescing optimizations using multiple strategies:

1. **Duplicate Removal:** Eliminates identical operations
2. **Contiguous Coalescing:** Merges operations accessing adjacent memory regions
3. **Non-blocking Batching:** Optimizes scheduling of non-blocking operations

3 Algorithm

4 Implementation Details

4.1 Operation Coverage

The pass supports all major OpenSHMEM RMA operations:

- **Memory operations:** `putmem`, `getmem`, `putmem_nbi`, `getmem_nbi`

Algorithm 1 OpenSHMEM Message Aggregation

Require: Function F containing OpenSHMEM operations

Ensure: Optimized function with coalesced operations

```
1:  $accesses \leftarrow \emptyset$ 
2:  $groups \leftarrow \emptyset$ 
   {Phase 1: Extract Memory Access Patterns}
3: for each operation  $op$  in  $F$  do
4:   if  $op$  is OpenSHMEM RMA operation then
5:      $access \leftarrow \text{EXTRACTMEMORYACCESS}(op)$ 
6:      $accesses \leftarrow accesses \cup \{access\}$ 
7:   end if
8: end for
   {Phase 2: Group Compatible Operations}
9: for each  $access$  in  $accesses$  do
10:   $key \leftarrow (access.pe, access.ctx, access.isNonBlocking,$ 
11:     $access.opType, access.opVariant, access.elementSize)$ 
12:  if  $groups[key]$  exists then
13:     $groups[key].operations \leftarrow groups[key].operations \cup \{access.op\}$ 
14:  else
15:     $groups[key] \leftarrow \text{CREATEGROUP}(access)$ 
16:  end if
17: end for
   {Phase 3: Apply Coalescing Transformations}
18: for each  $group$  in  $groups$  where  $|group.operations| \geq 2$  do
19:   if  $\text{AREALLOPERATIONSIDENTICAL}(group)$  then
20:      $\text{REMOVEDUPLICATES}(group)$ 
21:   else if  $\text{CANCREATECOALESCEDOPERATION}(group)$  then
22:      $\text{CREATECOALESCEDOPERATION}(group)$ 
23:   else if  $group.isNonBlocking$  and  $\text{CANBATCH}(group)$  then
24:      $\text{BATCHOPERATIONS}(group)$ 
25:   end if
26: end for
27: return optimized function  $F'$ 
```

- **Typed operations:** `put`, `get`, `put_nbi`, `get_nbi`
- **Context-aware variants:** `ctx_put`, `ctx_get`, `ctx_put_nbi`, `ctx_get_nbi`
- **Sized operations:** `put8/16/32/64/128`, `get8/16/32/64/128`
- **Point operations:** `p`, `g`

4.2 Safety Constraints

The transformation engine ensures correctness through several safety mechanisms:

1. **Memory Safety:** Operations are only coalesced if they access compatible memory regions
2. **Ordering Preservation:** OpenSHMEM ordering semantics within contexts are maintained
3. **Type Compatibility:** Only operations with identical types and variants are combined
4. **Context Isolation:** Operations using different contexts are never coalesced

4.3 Configuration Options

The pass provides several configuration parameters:

- `enablePutCoalescing`: Enable/disable PUT operation coalescing
- `enableGetCoalescing`: Enable/disable GET operation coalescing
- `minMessageSize`: Minimum message size threshold for coalescing
- `maxCoalescingDistance`: Maximum distance between coalescable operations
- `debugMode`: Enable detailed debug output

5 Experimental Results

The pass was evaluated on a comprehensive test suite of 14 test cases covering various OpenSHMEM operation patterns. Key results include:

- **Duplicate Elimination:** Successfully removes redundant identical operations
- **Type Safety:** Correctly prevents coalescing of incompatible operation types (e.g., `put32` vs `put64`)

- **Context Awareness:** Properly handles context-based operation separation
- **Blocking Behavior:** Maintains distinction between blocking and non-blocking operations

6 Future Work

The current implementation provides a solid foundation for advanced optimizations:

1. **Contiguous Memory Coalescing:** Merge operations accessing adjacent memory regions into single larger transfers
2. **Cross-operation Type Optimization:** Combine multiple small operations into larger sized operations (e.g., `put32 + put32 → put64`)
3. **Synchronization Boundary Analysis:** Optimize across synchronization points while preserving correctness
4. **Performance Heuristics:** Implement cost models to guide transformation decisions

7 Conclusion

The OpenSHMEM Message Aggregation Pass demonstrates the effectiveness of MLIR’s pattern rewriting infrastructure for domain-specific optimizations. By providing a comprehensive framework for analyzing and optimizing OpenSHMEM communication patterns, the pass enables significant performance improvements while maintaining correctness guarantees. The modular architecture and extensive test coverage provide a robust foundation for future research in HPC compiler optimizations.